Abductive Accumulation: Why Interpretive Errors Are Not Eliminated, Only Suppressed

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Abstract

The correction model of abduction holds that erroneous hypotheses are detected and replaced. We argue that this model is structurally inadequate in a precise sense: prior interpretive frames are not overwritten but *archived*. When a cognitive agent adopts a new hypothesis, the prior hypothesis is not eliminated; it is demoted to a latent layer that remains accessible and may be reactivated under appropriate conditions. What changes is not the presence of the prior frame but the *selection rule* that determines which frame is foregrounded. We call this process *abductive accumulation* and the resulting representational structure a *layered interpretive stack*. This account explains why abductive error is not correctable: the error is not a missing piece but a latent frame competing for selection. The practical consequence is that reliable cognition requires not the ability to delete erroneous frames—which is impossible—but the ability to suppress their reactivation: a *selection stability* condition distinct from but related to the negative abductive capability introduced in prior work [**?**]. We formalize the layered interpretive stack, the selection rule, and the conditions under which suppression fails, and derive implications for the understanding of cognitive bias, clinical reasoning errors, and the design of trustworthy AI systems.

Contents

1 Introduction

The correction assumption. The standard model of abduction—inference to the best explanation [**?**]—presents hypothesis revision as correction: a prior hypothesis h_t is found inadequate, and a new hypothesis h_{t+1} is installed in its place. The prior hypothesis, on this picture, is simply gone. The agent's representational state after revision is characterized by h_{t+1} alone; h_t plays no further role.

The residue phenomenon. This picture conflicts with a well-documented class of phenomena. Human reasoners who have been corrected on a factual belief nonetheless revert to the prior belief under time pressure, emotional arousal, or situations structurally similar to those in which the prior belief was formed [**?**]. Eyewitness testimony is unreliable not because new information is unavailable but because prior perceptual frames compete with later corrective accounts [**?**]. Anchoring effects persist after anchor correction has been explicitly performed [**?**]. In each case, the erroneous prior frame is not absent—it is *latent and reactivatable.*

The central claim. We argue that this class of phenomena reflects a structural feature of abductive cognition: interpretive frames are not overwritten but *archived.* The representational state of a cognitive agent is not a single active hypothesis but a *layered interpretive stack*—an ordered collection of frames, each associated with the evidence conditions under which it was formed. When a new frame is adopted, the prior frame is not deleted; it is pushed down the stack. What changes is the *selection rule*: the function that determines, given current context, which frame is foregrounded for use in perception, classification, and inference.

Consequence for the theory of abductive error. If prior frames are archived rather than deleted, then abductive error is not correctable in the sense of the correction model. A corrective intervention that installs h_{t+1} does not eliminate h_t; it adds h_{t+1} to the stack and modifies the selection rule. Under conditions that activate the selection criteria associated with h_t—time pressure, high similarity to original encoding context, emotional state, attentional load—h_t may be foregrounded again. The error persists as a latent selection risk.

Relation to prior work. This paper is the fourth in a series. Paper 1 [**?**] formalizes symbol grounding as a topological problem. Paper 2 [**?**] establishes hardware implementation conditions for grounding. Paper 3 [**?**] introduces the overwrite model of abduction and negative abductive capability (NAC). The present paper refines Paper 3: the overwrite model is a first approximation in which the prior frame is treated as eliminated upon transition. We now show that the prior frame is in fact archived, and that this archival structure is the structural basis of abductive non-correctability.

Remark 1 (Relation to Paper 3: refinement, not refutation)**.** *Paper 3's overwrite model correctly describes the* selection rule change *that occurs at commitment time t^*: a new interpretive frame F_{t+1} is installed as the default-active frame, and the agent's subsequent behavior is governed by F_{t+1} rather than F_t. This description remains accurate. The present paper adds a structural claim about what happens to F_t after the overwrite: rather than being deleted, F_t is archived in the interpretive stack and remains selectable under appropriate environmental conditions. Paper 3 provides the pre-commitment account (NAC as the capacity to withhold premature overwrite); the present paper provides the post-commitment account (SSC as the capacity to suppress reactivation of archived frames). The two papers are therefore complementary descriptions of different phases of the same process, not competing accounts.*

Central thesis. *Abductive error is not fully correctable because prior interpretive frames are not overwritten but archived; what changes is only the selection rule. Reliable cognition therefore requires not the ability to delete erroneous frames but the ability to suppress their reactivation.*

2 The Layered Interpretive Stack

2.1 From Single Frame to Stack

In Paper 3 [**?**], the representational state of an agent was modeled as a single interpretive frame $F(R)$. An overwrite transition $R_t \to R_{t+1}$ was defined as a state change in which $F(R_{t+1}) \not\approx F(R_t)$.

We now replace the single-frame model with a richer structure.

Definition 1 (Interpretive Stack)**.** *An interpretive stack $\mathcal{L} = (F_1, F_2, \ldots, F_n)$ is an ordered sequence of interpretive frames, where F_1 is the most recently adopted frame (top of stack) and F_n is the earliest (bottom of stack). Each frame F_k is associated with an* acquisition context *C_k: the set of evidence conditions, attentional states, and temporal markers under which F_k was formed.*

Definition 2 (Selection Rule)**.** *A selection rule $\mathcal{S} : \mathcal{E} \to \{1, \ldots, n\}$ maps the current environmental state $e \in \mathcal{E}$—including evidence quality, time pressure, emotional arousal, and contextual similarity to prior acquisition contexts—to an index k, selecting frame F_k as the active interpretive frame for that moment. A canonical functional form is:*

$$\mathcal{S}(e) = \arg\max_k \ \mathrm{softmax}_k\left(\frac{s(e,\, C_k)}{\tau}\right)$$

where C_k is the acquisition context of F_k, $s(\cdot,\cdot) : \mathcal{E} \times \mathcal{E} \to [0,1]$ is a context-similarity function, and $\tau > 0$ is a temperature parameter controlling selection sharpness. High τ produces near-uniform selection (reactivation risk increases); low τ produces sharp selection of the contextually best-matching frame.

Remark 2 (The temperature parameter τ and cognitive load)**.** *The temperature parameter τ has a concrete cognitive interpretation. Under normal operating conditions—attentive processing, low time pressure, familiar but non-overlapping context—τ is small and selection is sharp: F_1 (the top-of-stack corrected frame) dominates with high probability. As cognitive load increases, τ rises and the selection distribution flattens, increasing the probability that archived frames F_k ($k > 1$) are activated. Concretely, in LLMs: very long contexts require integrating over more tokens, effectively raising τ for context-similarity estimation; high-perplexity inputs lie far from the fine-tuning distribution, degrading $\delta(\cdot, C_1)$ and shifting probability toward earlier acquisition contexts; removal of chain-of-thought scaffolding removes the structured cue that normally suppresses F_k activation. In biological agents: time pressure, emotional arousal, and divided attention correspond directly to elevated τ (degraded selection rule application), producing the load-dependent reactivation effect documented in the continued influence literature [**?**]. This mapping from τ to observable cognitive conditions*

makes Prediction 3 (Load-Dependent Reactivation) directly testable: any experimental manipulation that increases τ should increase archived-frame reactivation rate.

Remark 3. *The correction model corresponds to the degenerate case in which the stack has depth 1 at all times: each overwrite replaces the stack with a singleton containing only F_{t+1}. We deny this degeneracy.*

2.2 Abductive Accumulation

Definition 3 (Abductive Accumulation)**.** *An abductive transition is an* accumulation *rather than an overwrite if the prior frame F_t is archived (pushed onto the stack) rather than deleted when the new frame F_{t+1} is adopted. The stack after accumulation is $(F_{t+1}, F_t, \ldots)$.*

Proposition 1 (Non-Correctability and Reactivation Bound)**.** *Let $\mathcal{L} = (F_{t+1}, F_t, \ldots)$ be the interpretive stack after a corrective accumulation at time $t+1$. Then:*

(i) F_t remains present in $\mathcal{L}$ and can be selected by $\mathcal{S}$ under environmental conditions e such that $\mathcal{S}(e) = k$ where $F_k = F_t$. The "correction" does not eliminate F_t; it installs F_{t+1} as the default-selected frame and modifies the selection rule.

(ii) The reactivation probability of F_t is bounded away from zero for all inputs e (since $\mathrm{softmax}(\cdot) > 0$ always), and is highest when $s(e, C_t)$ is large—that is, when e closely resembles the acquisition context C_t of F_t. Formally, $P(\mathcal{S}(e) = k \mid F_k = F_t) > 0$ for all e, with P increasing monotonically in $s(e, C_t)$. This bound cannot be driven to zero by modifying F_{t+1} alone, since $\mathcal{S}$ responds to $s(e, C_t)$, not to the content of F_{t+1}.

Under conditions that activate F_t's acquisition context C_t, F_t may be foregrounded, reinstating the prior bias. The agent's ability to prevent this reactivation defines Selection Stability Capacity (SSC), introduced in Section 3.

Corollary 1. *Abductive error is not a deleted state. It is a latent frame with a non-zero selection probability under a subset of environmental conditions. "Correcting" an abductive error reduces but does not eliminate this selection probability.*

3 Selection Instability as the Source of Bias

Definition 4 (Selection Stability)**.** *A cognitive agent exhibits* selection stability *with respect to hypothesis h^* if the selection rule $\mathcal{S}$ assigns h^*'s frame to the top index across all relevant environmental conditions $e \in \mathcal{E}$: $\mathcal{S}(e) = 1$ for all $e \in \mathcal{E}$.*

Proposition 2 (Bias as Selection Instability)**.** *Cognitive bias is not the presence of an erroneous frame in the stack but the failure of selection stability: the erroneous frame F_{err} is selected under a non-negligible subset of environmental conditions, producing systematic distortion of perception, classification, or inference under those conditions.*

Remark 4. *This reformulation has a direct implication for debiasing interventions. Interventions that provide corrective information—installing a new frame F^* at the top of the stack—are insufficient if they do not also modify the selection rule to suppress F_{err} reactivation. The corrective information may be present in the stack and yet fail to be selected under the conditions that matter most.*

4 Relation to Negative Abductive Capability

In Paper 3 [?], negative abductive capability (NAC) was defined as the capacity to maintain a suspension state—holding competing hypotheses without committing to any single frame—until evidence warrants commitment.

The present account adds a new dimension. NAC, as defined in Paper 3, addresses the *pre-commitment* phase: it prevents premature stacking. The present paper identifies a complementary *post-commitment* capacity: the ability to suppress reactivation of archived frames after commitment has occurred.

Definition 5 (Selection Stability Capacity (SSC))**.** *An agent possesses* selection stability capacity *if it can maintain $\mathcal{S}(e) = 1$ (top-of-stack selection) across the range of environmental conditions $\mathcal{E}$ relevant to a target task, even when the stack contains archived frames with overlapping acquisition contexts.*

Remark 5. *NAC and SSC are distinct but related capacities. NAC prevents premature accumulation of frames with high error risk. SSC prevents reactivation of accumulated frames with high error risk. Together they constitute the structural conditions for reliable abductive cognition:*

(NAC) *Suspend commitment until evidence warrants stacking.*

(SSC) *Once committed, suppress reactivation of archived error frames under conditions that would otherwise trigger them.*

5 Evidence and Illustrations

5.1 The Continued Influence Effect

The continued influence effect [?] refers to the persistence of misinformation in reasoning even after explicit correction. Subjects who are told that a belief was false continue to reason as if it were true, particularly under cognitive load.

On the accumulation model: the corrective information installs a new frame F^* at the top of the stack, but the prior frame F_{err} is archived with acquisition context C_{err}. Under cognitive load—which reduces the computational resources available for selection rule application—the selection process defaults to simpler criteria, reactivating F_{err}.

5.2 Anchoring and Adjustment

Anchoring effects [?] arise when an initial numerical estimate biases subsequent judgment even after adjustment information is provided.

On the accumulation model: the anchor establishes F_{anchor} as the initial frame. Adjustment installs F_{adj} but archives F_{anchor}. Insufficient adjustment reflects incomplete modification of the selection rule: $\mathcal{S}$ continues to assign partial weight to F_{anchor} under conditions similar to the original anchoring context.

5.3 *C. elegans*: Minimal Stack Depth

The predict-and-correct chemotaxis loop of *C. elegans* [**?**] may implement a minimal stack of depth 2: the current directional hypothesis and the most recent prior. The selection rule is implemented in the gain dynamics of the gradient detection circuit: when current evidence is strong, the top frame is selected; when evidence is weak or contradictory, the prior frame may contribute.

This would explain why *C. elegans* exhibits partial reversion to prior trajectories under gradient reversal conditions—a prediction of the accumulation model that distinguishes it from the pure overwrite model.

6 Implications

6.1 For Cognitive Science

The accumulation model predicts that:

1. Debiasing interventions that provide only corrective information will show incomplete and context-sensitive effects.

2. The magnitude of bias reactivation will correlate with the similarity between current environmental conditions and the original acquisition context C_{err}.

3. High cognitive load, time pressure, and emotional arousal will systematically increase reactivation probability by degrading selection rule application.

6.2 For AI Systems

In artificial systems, the accumulation model predicts:

1. Fine-tuning that installs a corrective frame F^* does not eliminate the prior frame F_{err} from the weight space; it modifies the selection dynamics.

2. Under distribution shift conditions that resemble the original training distribution, the prior frame may be preferentially activated.

3. Reliable debiasing requires not only corrective training but modification of the selection mechanism—e.g., via attention reweighting or gating mechanisms that suppress prior-context activation.

6.3 Connection to d_{cog}

In the terms of [**?**], the layered interpretive stack implies that the grounding-induced topology τ_A of an agent is not a single topology but a *mixture*: a weighted combination of the topologies induced by each frame in the stack, with weights determined by the selection rule. Under the persistence-signature formulation of [**?**, Def. **??**], the cognitive distance $d_{\text{cog}}(A, B)$ between agents with different stack histories decomposes into components that track distinct aspects

of this mixture: the d_{GH} component reflects worst-case distortion of the currently-selected frame, while the $d_{\mathrm{bn}}^{(k)}$ components reflect persistence-level alignment that integrates over the active stack region. Differences arising from divergent stack histories therefore show up as $d_{\mathrm{bn}}^{(k)}$ differences that would be invisible to a scalar GH measurement — a concrete point of contact between the accumulation model of this paper and the measurement framework of Paper 1.

More explicitly, a companion experiment reported in [**?**, App. **??**] exhibits a closed-form relationship between the geometry of an agent's interpretive-frame H_1 cycle and its persistence lifetime: under a constructed metric setting, the lifetime is $1-\sqrt{1-\cos g(\Theta)}$, where g is the maximal adjacent gap in the cycle. If the accumulation model is correct, stack histories that install new frames on top of old ones modify $g(\Theta)$ through the position of the newly-installed concept in the ring structure of the previously-dominant frame. Whether the new frame overwrites a critical node or a redundantly-placed one determines whether the prior frame's H_1 feature survives or collapses. This gives the accumulation/overwrite distinction a numerical signature that neither paper could supply in isolation from the measurement framework of Paper 1.

The persistence signature is therefore not a dynamic extension of d_{cog} in the strong sense of being time-indexed; it is a *structurally richer static snapshot* whose variation across stack-history-differing agents captures the accumulation-specific phenomena this paper addresses. A fully time-indexed extension (e.g. via zigzag persistence) is left to future work.

7 Pilot Empirical Evidence

The accumulation model predicts that prior representational structure persists after training rather than being deleted. We report a pilot study providing direct structural evidence for this claim in language models.

Setup. We applied RSA to Pythia-70m [**?**] at three training stages: step 0 (random initialization), step 1000 (early training), step 143000 (reference). Pairwise concept dissimilarity matrices were computed over $n = 29$ probe concepts at hidden layers $\ell \in \{2, 4, 6\}$, averaged over three prompt templates. Structural alignment was measured via second-order Spearman correlation ρ_{cog}.

Results. Three findings bear directly on the accumulation model.

Finding 1: Monotone structural increase. ρ_{cog} increases monotonically at all layers (L6: step-0 = 0.017; step-1000 = 0.535). If training were pure overwrite, step-0 and step-1000 would share no structural similarity with the reference beyond noise. The monotone increase is consistent with the accumulation model: prior structure is retained and built upon.

Finding 2: Non-zero baseline. Step-0 (random initialization) shows positive but weak correlation with the trained reference at all layers. Under the accumulation model, this reflects residual structural contribution of pre-training initialization to the final stack. Under the correction model, no such similarity is expected.

Finding 3: Cross-architecture convergence. $\rho_{\text{cog}}(\text{GPT-2}, \text{Pythia-trained}) \approx 0.42$ versus $\rho_{\text{cog}}(\text{GPT-2}, \text{Pythia-step-0}) \approx 0.10$ across all layers. Two architectures trained on different corpora converge to similar relational geometry, consistent with the accumulation model's prediction that training installs distribution-determined frames.

Interpretation. These results are consistent with the subsection on AI systems (Section 6.2): fine-tuning modifies selection dynamics without eliminating prior frame structure. Finding 2 provides an initial quantitative handle on "selection instability as the source of bias" (Section 3): the residual structure of step-0 in the trained reference is precisely the accumulated prior that can be reactivated under conditions resembling the original acquisition context. Direct mechanistic tests (circuit persistence, context-dependent reactivation) remain as future work. All code is available at `https://github.com/minamominamoto/grounding-without-reality`.

8 Open Problems

OP1. Formal characterization of the selection rule. What is the computational form of $\mathcal{S}$? Candidate: a softmax over frame-context similarity scores, where similarity is computed between current environmental state and each frame's acquisition context.

OP2. Stack depth in biological systems. What is the effective stack depth in human cognition? In *C. elegans*? Is there a minimum depth required for flexible behavior?

OP3. Measurement of reactivation probability. How can the selection probability of an archived frame be measured empirically? Candidate: vary environmental conditions systematically and measure behavioral or neural signatures of frame reactivation.

OP4. SSC and NAC joint optimization. What is the relationship between NAC (pre-commitment suspension) and SSC (post-commitment suppression)? Is there a tradeoff: systems that maintain longer suspension may have shallower stacks and therefore lower reactivation risk?

OP5. Fine-tuning as accumulation. Does fine-tuning in large language models implement accumulation rather than correction? If so, what are the signatures of prior-frame reactivation in fine-tuned models, and how can they be measured via mechanistic interpretability tools [**?**]?

9 Conclusion

The correction model of abduction is wrong in a precise and consequential way: it assumes that prior interpretive frames are deleted upon revision. They are not. They are archived in a layered interpretive stack, where they remain as latent selection risks.

What we call "correcting" an abductive error is more accurately described as installing a new top-of-stack frame and modifying the selection rule. The error persists; what changes is its probability of being foregrounded.

This has a direct implication for the theory of bias. Bias is not the presence of an erroneous frame. It is selection instability: the erroneous frame being selected under conditions where the correct frame should dominate. And it has a direct implication for the design of reliable cognitive systems, biological or artificial: the relevant capacity is not error deletion (impossible) but selection stability under adversarial conditions (achievable in principle).

Abductive errors are not eliminated by correction.
They persist as latent frames. They can only be suppressed.

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